

NS-R Formalism: $\{x^m, \chi^m\}, m=0, 1, \dots, D-1$

* χ_A^m is a D -dim vector i.e. it has an index in $\mathbb{M}^D$ and Spin index $A: \text{Spin}(2)$

$\rightarrow$ Theory has $(1,1)$ Susy:

Susy Recap: $N=(2,2), (x^0, x^1, \theta^\pm, \bar{\theta}^\pm)$

Chiral Superfield: $\bar{D}_\pm \Phi = 0$

• We obtain $N=(1,1)$ by restricting to a Subspace of $N=(2,2)$

$N=(1,1)$: θ^+ and θ^- are real up to a phase

$$\theta^+ = i e^{i\psi_+} \theta_2^+, \quad \theta_1^+ : \text{real}$$

$$\theta^- = i e^{i\psi_-} \theta_1^-, \quad \theta_2^- : \text{real}$$

$$\Phi = \phi + i\theta_1^+ \psi_+ + i\theta_2^- \psi_- + i\theta_1^+ \theta_2^- f$$

$$+ : (1, \bullet) \Leftrightarrow \theta^+ \oplus \bar{\theta}^+ \Rightarrow (2, \bullet)$$

$$- : (\bullet, 1) \quad \theta^- \oplus \bar{\theta}^- \Rightarrow (\bullet, 2)$$

$\therefore$ Let us take, $D_\pm = -i \frac{\partial}{\partial \theta^\pm} - 2\theta^\pm \partial_\pm$

$$\int d^2x \int d\theta^+ d\theta^- (D_- \Phi^\mu D_+ \Phi^\nu) M_{\mu\nu}$$

$\sim$ This is (roughly) where the susy action may come from...

$$\int d^2x \left(-\frac{1}{2} \partial_\alpha x^\mu \partial^\alpha x_\mu - \frac{i}{2} \bar{\chi}^\mu \not{\partial} \chi^\nu + \frac{1}{2} N^\mu N^\nu \right) M_{\mu\nu}$$

$\rightarrow$ Couple this to sugra:

$$(1/\sqrt{4\pi} \chi^\mu N^\mu) \oplus (e^a \psi_a)$$

$$S = \frac{1}{2\alpha'} \int d^2 \xi e \left\{ -\frac{1}{2} e_a^\alpha \partial_\alpha X^\mu e^{\alpha\beta} \partial_\beta X^\mu - \frac{i}{2} \bar{\chi}^\mu \partial \chi^\mu + \frac{1}{2} N^\mu N^\mu + \frac{i\kappa}{2} \bar{\psi}_\alpha \partial X^\mu \gamma^\mu \chi^\mu - \frac{\kappa^2}{16} \bar{\psi}_\alpha \gamma^\beta \gamma^\mu \psi_\beta \bar{\chi}^\mu \chi^\mu \right\} \eta_{\mu\nu}$$

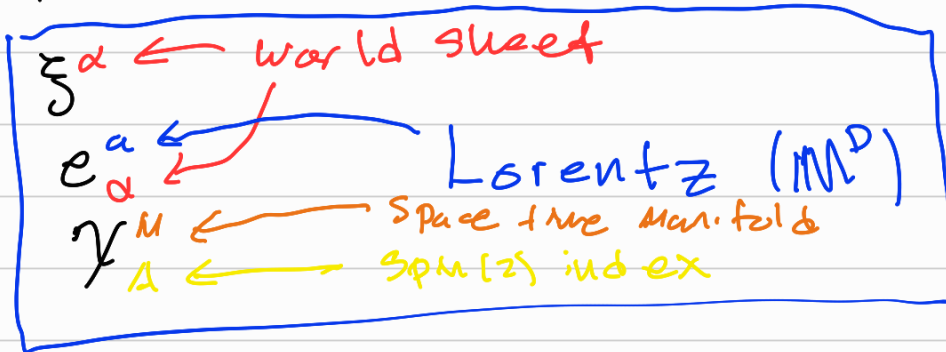
Symmetries:

(i) Diffeomorphism invariance:

$$\delta e_a^\alpha = f^B \partial_B e_a^\alpha + \partial_\alpha f^B e_B^\alpha, \quad \delta X^\mu = f^B \partial_B X^\mu$$

$$\delta \psi_{\alpha A} = f^B \partial_B \psi_{\alpha A} + \partial_\alpha f^B \psi_{BA}$$

$$\delta \chi_A^\mu = f^B \partial_B \chi_A^\mu, \quad \delta N^\mu = f^B \partial_B N^\mu$$



* Thus, $(X^\mu, N^\mu, \chi_A^\mu)$ are all treated as scalars

$$X^\mu(\xi^\alpha) \rightarrow X'^\mu(\xi'^\alpha) = X^\mu(\xi^\alpha) + f^B \partial_B X^\mu$$

$$\therefore \delta X^\mu = f^B \partial_B X^\mu$$

• Now, the fields with an α index transform as vectors:

$$\delta \psi^\alpha = f^B \partial_B \psi^\alpha$$

(ii) Local two-dim Lorentz invariance:

→ Recall Lorentz acts on 'a' index AS WELL
 AS the Spin(2) index "A".

$$\delta e_a^\alpha = -\omega^a_b e^\alpha_b$$

$$\delta \chi_A^\mu = \frac{1}{4} (\omega^{ab} \gamma_{ab})_A^B \chi_B^\mu$$

$$\delta \psi_{\alpha A} = \frac{1}{4} (\omega^{ab} \gamma_{ab})_A^B \psi_{\alpha B}$$

(iii) ~~Local~~ two-dim supersymmetry:

$$\begin{aligned}\delta x^m &= i\bar{\epsilon} \chi^m & \delta \chi^m &= (\hat{D} x^m + N^m) \epsilon \\ \delta N^m &= i\bar{\epsilon} \hat{D}' \chi^m \\ \delta e_a^\alpha &= i\kappa \bar{\epsilon} \gamma^a \psi_a, & \delta \psi_a &= \frac{2}{\kappa} D_a \epsilon\end{aligned}$$

$$\tilde{D}_\alpha \chi^m = \partial_\alpha \chi^m - \frac{i\kappa}{2} \Phi_\alpha \chi^m$$

$$\hat{D}_\beta \chi^m = (D_\beta \chi^m - \frac{\kappa}{2} \hat{D} \chi^m + N^m) \psi_\beta$$

$$D_\beta \chi^m = (\partial_\beta + \frac{1}{4} \omega_{\beta ab} \gamma^{ab}) \chi^m$$

$$D_\beta \epsilon = (\partial_\beta + \frac{1}{4} \omega_{\beta ab} \gamma^{ab}) \epsilon$$

(iv) Weyl-symmetry: